

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture(s):

- The sets A and B have the same cardinality if and only if there is a bijection $f : A \rightarrow B$.
- An infinite set A is countable if and only if there is a bijection $f : A \rightarrow \mathbb{Z}^+$
- $\mathbb{Z}$ and $\mathbb{Q}$ are countable.
- The real numbers in the interval $(0, 1)$ is uncountable.
- Any subset of a countable set is countable.
- If A and B are countable, then $A \cup B$ is also countable.

Sec 2.6 Matrices

Definition

- A matrix is a rectangular array of numbers.
- A matrix with m rows and n columns is called an $m \times n$ matrix. Notation:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \begin{matrix} 1 \leq i \leq m \\ 1 \leq j \leq n \end{matrix} \quad A = [a_{ij}]$$

- A is called a square matrix if $m = n$.

- The i th row is $[a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$, and the j th column is $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{bmatrix}$.

Matrix operations

- Two matrices are equal if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.
- Given two $m \times n$ matrices $A = [a_{ij}]$ and $B = [b_{ij}]$:

$$A + B = C,$$

where $c_{ij} = \underline{a_{ij}} + \underline{b_{ij}}$.

Matrix operations

- Given $m \times k$ matrix $A = [a_{ij}]$ and $k \times n$ matrix $B = [b_{ij}]$, the product

$$C = AB$$

is defined using

$$c_{ij} = \sum_{r=1}^k a_{ir} b_{rj}$$

Handwritten annotations: A red arrow points from the k in the first matrix to the upper limit of the summation. Another red arrow points from the k in the second matrix to the lower index r of the summation. Above the summation, the text $1 \dots m$ is written with an arrow pointing to the i index. To the right, the text $1 \dots n$ is written with an arrow pointing to the j index. A red bracket is drawn under the entire summation expression.

What is the size of C ?

$\rightarrow m \times n$

Example

Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -10 \\ 0 & 15 \end{bmatrix}$. Find AB .

3×2

2×2



$$\begin{bmatrix} 1 \cdot 4 + (-1) \cdot 0 & 1 \cdot (-10) + (-1) \cdot 15 \\ 0 \cdot 4 + 2 \cdot 0 & 0 \cdot (-10) + 2 \cdot 15 \\ 5 \cdot 4 + 3 \cdot 0 & 5 \cdot (-10) + 3 \cdot 15 \end{bmatrix} = \begin{bmatrix} 4 & -25 \\ 0 & 30 \\ 20 & -5 \end{bmatrix}$$

Let $A = \begin{bmatrix} 3 & 5 & 4 \\ 8 & 9 & 10 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Find AB .

2×3

3×3

2×3



$$\begin{bmatrix} 3 & 5 & 4 \\ 8 & 9 & 10 \end{bmatrix} = A$$

- The $n \times n$ identity matrix is

$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$

Matrices

Note: Given two square matrices A and B , we generally have

$$AB \neq BA$$

Example

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 1 & 6 \end{bmatrix}$$
$$AB = \begin{bmatrix} 1 & 6 \\ 3 & 9 \end{bmatrix} \quad BA = \begin{bmatrix} 3 & 5 \\ 6 & 7 \end{bmatrix}$$

Matrix operations

- Given an $n \times n$ matrix A and an integer k , define

$$A^0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^0 = I_n, \quad A^k = \underbrace{AAA \cdots A}_{k \text{ times}}$$

Example

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find A^2 and A^3 and A^n .

$$\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 0 & 1 \cdot 1 + 2 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

Matrix operations

- Given an $n \times n$ matrix A , the inverse of A is defined as a matrix B such that

$$AB = I_n$$

Such matrix need not exist. If it does exist, it is denoted by A^{-1} .

Example

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Show that its inverse exists if $ad \neq bc$. In this case,

$$A \cdot A^{-1} = I_n$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & -ab + ab \\ dc - dc & -bc + ad \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} \frac{1}{ad - bc} = I_n$$

Matrix operations

- Let $A = [a_{ij}]$ be an $m \times n$ matrix. The transpose of A , denoted by A^t , is an $n \times m$ matrix obtained by swapping the rows and columns of A : $A^t = [b_{ij}]$, where $b_{ij} = a_{ji}$.

Example 2×3

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Find A^t . 3×2

$$A^t = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Symmetric Matrix

Definition

A square matrix A is called symmetric if $A = A^t$.

Example

3x3

Is $A = \begin{bmatrix} a & -1 & 10 \\ -1 & b & 17 \\ 10 & 17 & c \end{bmatrix}$ symmetric? 

$$A^t = \begin{bmatrix} a & -1 & 10 \\ -1 & b & 17 \\ 10 & 17 & c \end{bmatrix} = A$$

Zero-One Matrices

- A matrix all of whose entries are either 0 or 1 is called a zero-one matrix.

Operations $\vee$ and $\wedge$:

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ zero-one matrices.

- The join of A and B is $C = A \vee B$, where $c_{ij} = a_{ij} \vee b_{ij}$.
- The meet of A and B is $C = A \wedge B$, where $c_{ij} = a_{ij} \wedge b_{ij}$.

Example

$$A = \begin{bmatrix} \underline{1} & 0 & 1 \\ \underline{0} & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} \underline{0} & 1 & 0 \\ \underline{1} & 1 & 0 \end{bmatrix}. \quad \underline{A \vee B} = \begin{bmatrix} \underline{1 \vee 0} & 0 \vee 1 & 1 \vee 0 \\ \underline{0 \vee 1} & 1 \vee 1 & \underline{0 \vee 0} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & \underline{0} \end{bmatrix}$$
$$A \wedge B = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Zero-One Matrices

Let $A = [a_{ij}]$ be an $m \times k$ zero-one matrix and $B = [b_{ij}]$ be a $k \times n$ zero-one matrix. Then the Boolean product of $C = A \odot B$, is the $m \times n$ matrix given by

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \cdots \vee (a_{ik} \wedge b_{kj}).$$

Example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

$$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 0) \vee (0 \wedge 1) & (1 \wedge 1) \vee (0 \wedge 0) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 0) \vee (1 \wedge 1) & (0 \wedge 1) \vee (1 \wedge 0) \\ (1 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 0) \vee (1 \wedge 1) & (1 \wedge 1) \vee (1 \wedge 0) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$